R and clinical/lab data

2026 edition

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Welcome

Welcome to this course, here you can find the notebooks used in the PhD course of 2026.

Chapter 1

Dataframes, vectors and numbers

1.1 Dataframes

The usual data you will deal with are tabular (rows x columns) data, usually improperly called "Excel files".

The standard formats for raw data files in data analyses are *.csv*, *.tsv* and *.txt* files; these should be our starting point.

1.2 Import file

The first thing to do is to import the file into R session. We can do it with several functions, but the standard one for *.csv* files is `read.csv`.

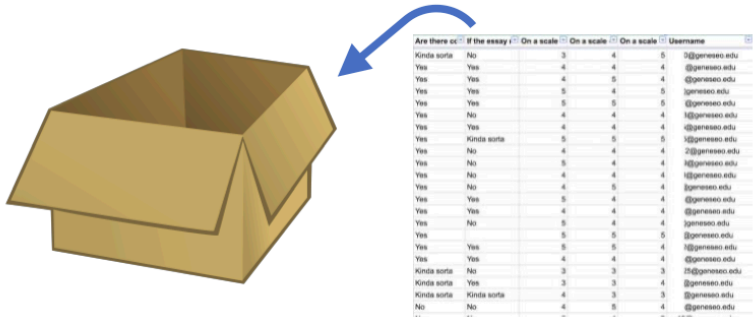
Let's see how to do so:

```
my_data <- read.csv("data/input_df.csv")
```

1.2.0.1 Notes about variables

Variables are the "memory" of an R session, they represent box where R stores data/values you want.

You can think them as boxes that have a name on them:



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Kinda sorta	No	3	4	5	3@geneseo.edu
Yes	Yes	4	4	4	@geneseo.edu
Yes	Yes	4	5	4	6@geneseo.edu
Yes	Yes	5	4	5	)geneseo.edu
Yes	Yes	5	5	5	6@geneseo.edu
Yes	No	4	4	4	1@geneseo.edu
Yes	Yes	4	4	4	4@geneseo.edu
Yes	Kinda sorta	5	5	5	1@geneseo.edu
Yes	No	4	4	4	2@geneseo.edu
Yes	No	5	4	4	8@geneseo.edu
Yes	No	4	4	4	1@geneseo.edu
Yes	No	4	5	4	)geneseo.edu
Yes	Yes	5	4	4	@geneseo.edu
Yes	Yes	4	4	4	@geneseo.edu
Yes	No	5	4	4	)geneseo.edu
Yes		5	5	5	)geneseo.edu
Yes	Yes	5	5	4	1@geneseo.edu
Yes	Yes	4	4	4	@geneseo.edu
Kinda sorta	No	3	3	3	75@geneseo.edu
Kinda sorta	Yes	3	3	4	)geneseo.edu
Kinda sorta	Kinda sorta	4	3	3	)geneseo.edu
No	No	4	5	4	@geneseo.edu

There are important **rules** about variables:

- In each variable you can store one single item (a number, a string, a dataframe, a list, ...)
- variable names can start only with a letter (uppercase or lowercase)
- variable names can contain **only** letters, numbers, underscores (_) and dots (.). **No spaces, no dash (-) or any other special characters**
- there could be only ONE variable with a specific name, saving an object with the same of another results in the substitution of the first one
- suggestion: variable names should be meaningful to you (and to the ones you share the code with): e.g. `x` is not a good name, while `patients_glucose_data` is preferred

1.2.1 First steps on a dataframe

When importing a dataframe, there is a fundamental step to do: *inspection*. Inspection helps us to understand how the dataframe is structured, which type

of data we have, which columns etc.

To do it, we use 3 functions that you need to remember (tatoo yourself): `summary`, `str` and `head`. Let's see them:

```
summary(my_data)
```

```
     sample        timepoint           genotype          treatment
 Min.   :  1.00   Length:192         Length:192         Length:192
 1st Qu.: 48.75   Class :character   Class :character   Class :character
 Median : 96.50   Mode  :character   Mode  :character   Mode  :character
 Mean   : 96.50
 3rd Qu.:144.25
 Max.   :192.00
      area        intensity
 Min.   :3000   Min.   : 9130286
 1st Qu.:3259   1st Qu.:16283904
 Median :3517   Median :18894104
 Mean   :3508   Mean   :19419175
 3rd Qu.:3769   3rd Qu.:22090201
 Max.   :3988   Max.   :29812114
```

```
str(my_data)
```

```
'data.frame':   192 obs. of  6 variables:
 $ sample   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ timepoint: chr  "P1" "P1" "P1" "P1" ...
 $ genotype : chr  "wt" "wt" "wt" "wt" ...
 $ treatment: chr  "n" "n" "n" "n" ...
 $ area     : num  3125 3500 3773 3538 3936 ...
 $ intensity: num  16015465 19449815 21118210 18100075 20718554 ...
```

We can see how many rows and columns we have, as well as which columns, the data type of each and the first entries.

Lastly, `head` is the function that shows us the first 6 entries of the dataframe, as a table:

```
head(my_data)
```

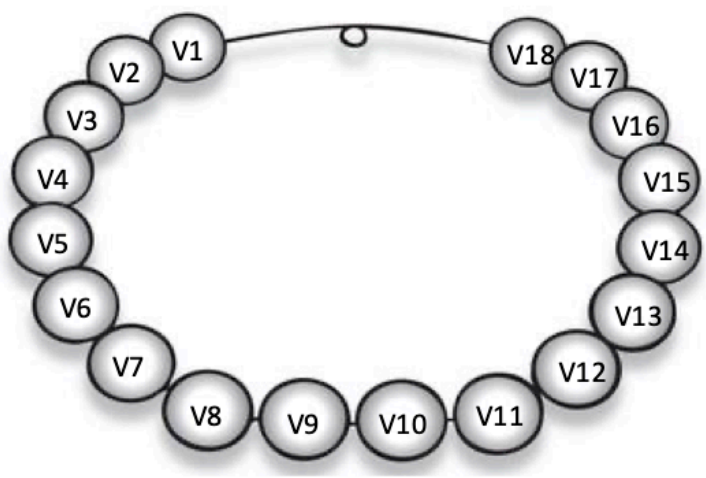
```
  sample timepoint genotype treatment     area intensity
1      1        P1       wt         n 3124.709  16015465
2      2        P1       wt         n 3499.559  19449815
3      3        P1       wt         n 3773.000  21118210
4      4        P1       wt         n 3538.320  18100075
```

```
5       5        P1       wt         n 3936.014  20718554
6       6        P1       wt         n 3646.389  19507110
```

1.3 Vectors

Dataframes are organized and read by columns. Each column is a **vector** in R.

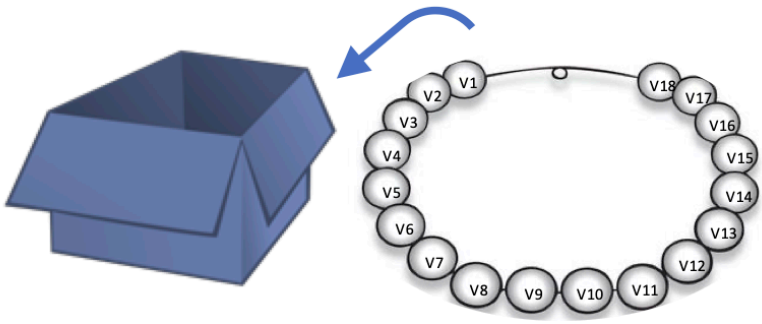
A vector is a collection of data of the **same type**; you can think of a vector as a pearl necklace, with each pearl being a single value:

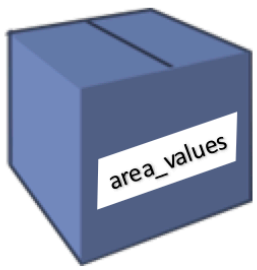


1.3.1 Accessing dataframe columns

For exercise purposes, we can create a new variable storing the values of the area column of the dataframe. To extract a single column from a dataframe, we use the structure `dataframe_name$column_name`.

```
area_values <- my_data$area
```





1.3.2 Accessing values of a vector: slicing

Slicing is the operation of extract values out of a variable (in this case a vector). It is done to answer these questions:

- *Which is the first value of the vector?*
- *Which is the value at position n of a vector?*
- *Which are the values from position n to m?*

There are different ways in which we can slice a vector, and today we will see indexing (through positions).

The syntax is:

- `variable_name[position]` to get one single value
- `variable_name[start:end]` to get elements from `start` to `end`

REMEMBER: In R, the first element is at index 1.

```
# extract one single value

first_area_value <- area_values[1]
print(first_area_value)
```

```
[1] 3124.709
```

```
# extract range of values
from_fifth_to_ninth_areas <- area_values[5:9]
print(from_fifth_to_ninth_areas)
```

```
[1] 3936.014 3646.389 3073.109 3717.063 3887.006
```

Q: *What 5:9 does in R?* A: It creates a vector containing numbers from 5 to 9, that's why it works:

```
print(5:9)
```

```
[1] 5 6 7 8 9
```

Q: *And to get last element?*

A: We need to know the lenght of the vector and use it as index. To get the length of a vector we use the function `length`:

```
length_area <- length(area_values)
print(length_area)
```

```
[1] 192
```

```
last_area <- area_values[length_area]
print(last_area)
```

```
[1] 3808.121
```

Exercise 1.1. Which is the 5th value of the intensity column of the dataframe?

Solution

We can do it in many steps:

```
intensity_values <- my_data$intensity

fifth_intensity <- intensity_values[5]

print(fifth_intensity)
```

```
[1] 20718554
```

Or in just one step:

```
fifth_intensity <- my_data$intensity[5]

print(fifth_intensity)
```

```
[1] 20718554
```

1.3.2.1 Replace a value by index

We can replace of a vector using indexes.

Note: new values should be of the same type (numbers for numbers, strings for strings ecc), otherwise everything will be changed to strings (most of the time).

To change a single value, we can type `variable_name[position] <- new_value`; to change multiple values `variable_name[start:end] <- new_values`.

1.3.3 Numbers in R

Now that we know how to get and extract numbers from numeric vectors, let's see what we can do with numbers in R.

1.3.3.1 Declare a new numeric variable

To declare a new numeric variable, we use the statement `variable_name <- number`:

```
conversion_factor <- 1000

print(conversion_factor)
```

```
[1] 1000
```

```r
print(typeof(conversion_factor))
```

```
[1] "double"
```

Decimal point numbers wants the dot `.` as decimal separator:

```r
decimal_n <- 215.6
print(decimal_n)
```

```
[1] 215.6
```

```r
print(typeof(decimal_n))
```

```
[1] "double"
```

1.3.3.2 Arithmetical operations

We can perform all sort of matematical operation:

```r
# sum

print(2 + 5)
```

```
[1] 7
```

```r
# subtraction

print(7 - 2.5)
```

```
[1] 4.5
```

```r
# multiplication
print(12 * 5)
```

```
[1] 60
```

```r
# division
print(105 / 43)
```

```
[1] 2.44186
```

```r
# exponential
print(5 ** 2)
```

```
[1] 25
```

```r
print(5 ^ 3)
```

```
[1] 125
```

```r
# square root
print(sqrt(12))
```

```
[1] 3.464102
```

We can combine them all to do more complex operations. For example, you can resolve this equation $x^2 - 7x + 12 = 0$. We know that the formula to resolve this equation is: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, so we can reconstruct them in R:

```r
a <- 1
b <- -7
c <- 12

x1 <- (-b + sqrt(b**2 - 4*a*c)) / (2 * a)
x2 <- (-b - sqrt(b**2 - 4*a*c)) / (2 * a)

print(x1)
```

```
[1] 4
```

```r
print(x2)
```

```
[1] 3
```

Exercise 1.2. Which are the last 3 values of the intensity column of the dataframe?

Solution

We can do it in many steps:

```
intensity_values <- my_data$intensity

length_intensity <- length(intensity_values)

third_to_last_index <- length_intensity - 2

last_3_intensities <- intensity_values[third_to_last_index:length_intensity]

print(last_3_intensities)
```

```
[1] 22965783 22027811 22338574
```

1.3.4 Vector-number operations

What we have seen so far on arithmetical operations between single numbers, can be applied also to numerical vector x single number operations.

In our data, area is expressed in µm^2 and we want to convert it into mm^2. To do so, we should divide each value of the area by a factor of 1,000,000; the cool thing about R is that this is done automatically when we use the statement `numeric_vector <operand> number`:

```
division_factor = 1e6

areas_mm2 <- area_values / division_factor

print(head(areas_mm2))
```

```
[1] 0.003124709 0.003499559 0.003773000 0.003538320 0.003936014 0.003646389
```

```
print(head(area_values))
```

```
[1] 3124.709 3499.559 3773.000 3538.320 3936.014 3646.389
```

This is true for ANY arithmetical operations:

```
print(head(intensity_values))
```

```
[1] 16015465 19449815 21118210 18100075 20718554 19507110
```

```
print(head(intensity_values + 10))
```

```
[1] 16015475 19449825 21118220 18100085 20718564 19507120
```

1.3.5 Operations between vectors

It is possible also to perform element-wise operations between vectors.

Let's load a new dataset with some patient data, and calculate the BMI of each patient.

```r
patient_df <- read.csv("data/patient_data.csv")

str(patient_df)
```

```
'data.frame':   100 obs. of  3 variables:
 $ patient_id: int  1 2 3 4 5 6 7 8 9 10 ...
 $ weight    : num  75.8 61.8 91.7 66.4 80.4 ...
 $ height    : num  186 212 188 176 174 ...
```

```r
summary(patient_df)
```

```
   patient_id         weight           height      
 Min.   :  1.00   Min.   : 49.79   Min.   : 91.88  
 1st Qu.: 25.75   1st Qu.: 68.25   1st Qu.:159.20  
 Median : 50.50   Median : 73.91   Median :173.15  
 Mean   : 50.50   Mean   : 75.18   Mean   :174.96  
 3rd Qu.: 75.25   3rd Qu.: 84.61   3rd Qu.:185.90  
 Max.   :100.00   Max.   :112.18   Max.   :253.13  
```

```r
head(patient_df)
```

```
  patient_id   weight   height
1          1 75.82265 185.6147
2          2 61.77137 212.2587
3          3 91.65377 188.3604
4          4 66.38117 176.2986
5          5 80.42689 174.0509
6          6 79.32015 173.7906
```

BMI formula: $\frac{\text{weight in kg}}{(\text{height in m})^2}$

To do so, we should transform the height in m, and then perform the operation. We will save the results as a **new column** of the dataframe; this can be done with the syntax `dataframe_variable$new_column_name <- vector` or `dataframe_variable$["new_column_name"] <- vector` (we will see on day 2 the strings).

```r
# Get height in m
patient_df$height_in_m <- patient_df$height / 100

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m
1          1 75.82265 185.6147    1.856147
2          2 61.77137 212.2587    2.122587
3          3 91.65377 188.3604    1.883604
4          4 66.38117 176.2986    1.762986
5          5 80.42689 174.0509    1.740509
6          6 79.32015 173.7906    1.737906
```

```r
patient_df["BMI"] <- patient_df$weight / (patient_df$height_in_m ** 2)

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI
1          1 75.82265 185.6147    1.856147 22.00768
2          2 61.77137 212.2587    2.122587 13.71059
3          3 91.65377 188.3604    1.883604 25.83279
4          4 66.38117 176.2986    1.762986 21.35733
5          5 80.42689 174.0509    1.740509 26.54902
6          6 79.32015 173.7906    1.737906 26.26220
```

1.3.6 Sumamry statistics of numeric vectors

Usually, when dealing with numeric data we want to have some summary statistics on a specific data (e.g. mean values, median, quartile, standard deviation, …).

In R there are many built-in functions that can help us in doing so:

- `mean` to calculate the mean
- `sd` to calculate the standard deviation
- `variance` to calculate the variance
- `median` to calculate the median
- `quantile` to calculate the nth quantile of a distribution
- `round` to round decimal values to n decimal places
- `sum` to sum all the values of the vector

- many others

They all have in common the syntax: `name_of_the_function(vector)`.

```
intensity_mean <- mean(intensity_values)
print(intensity_mean)
```

```
[1] 19419175
```

```
height_sd <- sd(patient_df$height)
print(height_sd)
```

```
[1] 26.73183
```

```
BMI_median <- median(patient_df$BMI)
print(BMI_median)
```

```
[1] 25.23683
```

```
first_quartile_weight <- quantile(patient_df$weight, probs = 0.25)
print(first_quartile_weight)
```

```
     25%
68.25118
```

```
rounded_BMI <- round(patient_df$BMI, 1)
print(head(rounded_BMI))
```

```
[1] 22.0 13.7 25.8 21.4 26.5 26.3
```

Exercise 1.3. Scale BMI values (formula: $scaled_i = \frac{x_i - \overline{x}}{sd(x)}$)

Solution

Here is a step-by-step solution:

```
# calculate BMI mean
mean_BMI <- mean(patient_df$BMI)

# calculate BMI sd
sd_BMI <- sd(patient_df$BMI)

# Calculate scaled BMIs
patient_df$scaled_BMI <- (patient_df$BMI - mean_BMI) / sd_BMI

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI  scaled_BMI
1          1 75.82265 185.6147    1.856147 22.00768 -0.44542706
2          2 61.77137 212.2587    2.122587 13.71059 -1.29224320
3          3 91.65377 188.3604    1.883604 25.83279 -0.05502959
4          4 66.38117 176.2986    1.762986 21.35733 -0.51180319
5          5 80.42689 174.0509    1.740509 26.54902  0.01807031
6          6 79.32015 173.7906    1.737906 26.26220 -0.01120341
```

That is exactly what the formula `scale` do underneath:

```r
# Calculate scaled BMIs
patient_df$scaled_BMI_function <- scale(patient_df$BMI)

print(head(patient_df))
```

```
  patient_id   weight   height height_in_m      BMI  scaled_BMI
1          1 75.82265 185.6147    1.856147 22.00768 -0.44542706
2          2 61.77137 212.2587    2.122587 13.71059 -1.29224320
3          3 91.65377 188.3604    1.883604 25.83279 -0.05502959
4          4 66.38117 176.2986    1.762986 21.35733 -0.51180319
5          5 80.42689 174.0509    1.740509 26.54902  0.01807031
6          6 79.32015 173.7906    1.737906 26.26220 -0.01120341
  scaled_BMI_function
1         -0.44542706
2         -1.29224320
3         -0.05502959
4         -0.51180319
5          0.01807031
6         -0.01120341
```

1.4 Save a dataframe

We are satisfied with this preliminary edit of the patients' data, so we can save the data to a dedicated file.

We use `write.csv` function:

```r
write.csv(patient_df, # what we want to save
          file = "output/patients_data_BMI.csv", # name of the file
          row.names = FALSE) # we'll see row names in day 2
```

1.5 Home exercise

For next time, if you want, you can try to do this exercise:

1. Starting from patient data (raw), load the file
2. Inspect it
3. You know that you have to give drug A to each patient so that the final concentration is 10 mg/kg (mg of drug every kg of patient). Calculate how much drug you should give to each patient.
4. Given that a single stock of drug A is 5 g, how many stocks you have to order?

Solution

```r
# 1. Load the file
patient_df <- read.csv("data/patient_data.csv")
```

```r
# 2. Inspect it
head(patient_df)
```

```
  patient_id   weight   height
1          1 75.82265 185.6147
2          2 61.77137 212.2587
3          3 91.65377 188.3604
4          4 66.38117 176.2986
5          5 80.42689 174.0509
6          6 79.32015 173.7906
```

```r
str(patient_df)
```

```
'data.frame':   100 obs. of  3 variables:
 $ patient_id: int  1 2 3 4 5 6 7 8 9 10 ...
 $ weight    : num  75.8 61.8 91.7 66.4 80.4 ...
 $ height    : num  186 212 188 176 174 ...
```

```r
summary(patient_df)
```

```
   patient_id         weight           height      
 Min.   :  1.00   Min.   : 49.79   Min.   : 91.88  
 1st Qu.: 25.75   1st Qu.: 68.25   1st Qu.:159.20  
 Median : 50.50   Median : 73.91   Median :173.15  
 Mean   : 50.50   Mean   : 75.18   Mean   :174.96  
 3rd Qu.: 75.25   3rd Qu.: 84.61   3rd Qu.:185.90  
 Max.   :100.00   Max.   :112.18   Max.   :253.13  
```

```r
# 3. Calculate drug for each patient

patient_df$drugA_quantity_mg <- patient_df$weight * 10
head(patient_df)
```

```
  patient_id   weight   height drugA_quantity_mg
1          1 75.82265 185.6147          758.2265
2          2 61.77137 212.2587          617.7137
3          3 91.65377 188.3604          916.5377
4          4 66.38117 176.2986          663.8117
5          5 80.42689 174.0509          804.2689
6          6 79.32015 173.7906          793.2015
```

```r
# 4. How many stocks you have to order?

total_drug_A_mg <- sum(patient_df$drugA_quantity_mg)

total_drug_A_g <- total_drug_A_mg / 1000

stock_weight <- 5

number_of_stocks <- total_drug_A_g/stock_weight

print(number_of_stocks)
```

```
[1] 15.03539
```

To be precise, we can use `ceiling` function that round to the upper integer:

```r
number_of_stocks <- ceiling(number_of_stocks)

print(number_of_stocks)
```

```
[1] 16
```